

AromaUI



Enterprise-grade development toolkit for cross-platform applications on Linux, Android, and embedded systems.

Version 0.0.1 (alpha)

Generated March 12, 2026

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Table of Contents

SDK Installation	3
Project Setup	4
Hello World Tutorial	5
Architecture Overview	6
Core JNI API	7
AromaABI Interface	8
System Services	9
Device Information	10
Bluetooth Classic API	11
Wi-Fi API	12
UI Utilities	13
Widgets Library	14
Layout Management	15
DPI Conversion	16
Orientation API	17
Theming System	18
Gestures & Touch	19
Camera API	20
File System API	21
Preferences API	22
Permissions API	23
Building & Deployment	24
Logging System	25

SDK Installation

1. Download the latest release

You can download the latest release of the software from our [GitHub releases page](#). Look for the most recent version and click on the download link to get the installer.

2. Run the installer

Once the download is complete, locate the installer file (usually in your Downloads folder) and double-click on it to run the installer. Follow the on-screen instructions to complete the installation process. You may be prompted to choose an installation location and agree to the terms and conditions. Make sure to read through the terms before accepting.

3. Start Developing

Check out the **Project Setup** documentation to learn how to create your first project with AromaUI and start building amazing user interfaces!

4. Verify Installation

To verify that the installation was successful, you can run the following command in your terminal or command prompt:

```
aroma doctor
```

This command will check if AromaUI is properly installed and configured on your system. If everything is set up correctly, you should see a message confirming that the installation is successful. If there are any issues, the command will provide details on what needs to be fixed.

```
yassine@DESKTOP-4SJS544:~/AromaUI$ aroma doctor
==> Running Aroma Doctor...
OS: Linux 6.6.87.2-microsoft-standard-WSL2
✓ CMake installed
✓ Ninja: 1.11.1
✓ GCC installed
✓ Java installed
✓ keytool installed
✓ Android SDK: /home/yassine/Android/Sdk
✓ Android NDK: /home/yassine/Android/Sdk/ndk/25.1.8937393
==> Doctor summary complete.
```

Project Setup

1. Project Creation via Command Line Tool

To create a new AromaUI project, you can use the `aroma` command-line tool. Open your terminal and run the following command:

```
aroma create
```

This command will prompt you to enter a project name and select the min, target and compile android API versions.

After providing the necessary information, the tool will generate a new project with the appropriate structure and configuration files.

2. Project Structure Overview

A typical AromaUI project contains:

```
project/  
├─ src/  
│   └─ main.c  
├─ android/  
├─ CMakeLists.txt  
└─ aroma.json
```

Your application logic is written in C inside `src/`.

Hello World Tutorial

Below is a minimal AromaUI example:

```
#include "aroma.h"

int main(int argc, char** argv)
{
    if (!aroma_ui_init()) {
        return 1;
    }

    AromaTheme theme = aroma_theme_create_material_black();
    aroma_ui_set_theme(&theme);

    AromaWindow* win = aroma_ui_create_window("Hello World!", 400, 800);
    aroma_window_set_fullscreen((AromaNode*)win, true);

    while (aroma_ui_is_running()) {
        aroma_ui_process_events();
        aroma_ui_render(win);
    }

    aroma_ui_shutdown();
    return 0;
}
```

For Android support, include the native entry point:

```
#ifdef __ANDROID__
#include <android_native_app_glue.h>
void android_main(struct android_app* state)
{
    aroma_android_set_app(state);
    main(0, NULL);
}
#endif
```

Explanation of the Code

- `aroma_ui_init()` : Initializes the AromaUI system. Must be called before any other UI functions.
- `aroma_theme_create_material_black()` : Creates a Material Design black theme.
- `aroma_ui_set_theme()` : Sets the current theme for the UI.
- `aroma_ui_create_window()` : Creates a new window with the specified title and dimensions
- `aroma_window_set_fullscreen()` : Sets the window to fullscreen mode.
- `aroma_ui_is_running()` : Checks if the UI is still running (i.e the window is open).
- `aroma_ui_process_events()` : Processes input events (e.g. keyboard, mouse, touch).
- `aroma_ui_render()` : Renders the UI to the specified window.
- `aroma_ui_shutdown()` : Cleans up and shuts down the AromaUI system.

Window dimensions and title are both ignored on Android, as the app will always inherit screen dimensions and use the app name from the manifest.

Architecture Overview

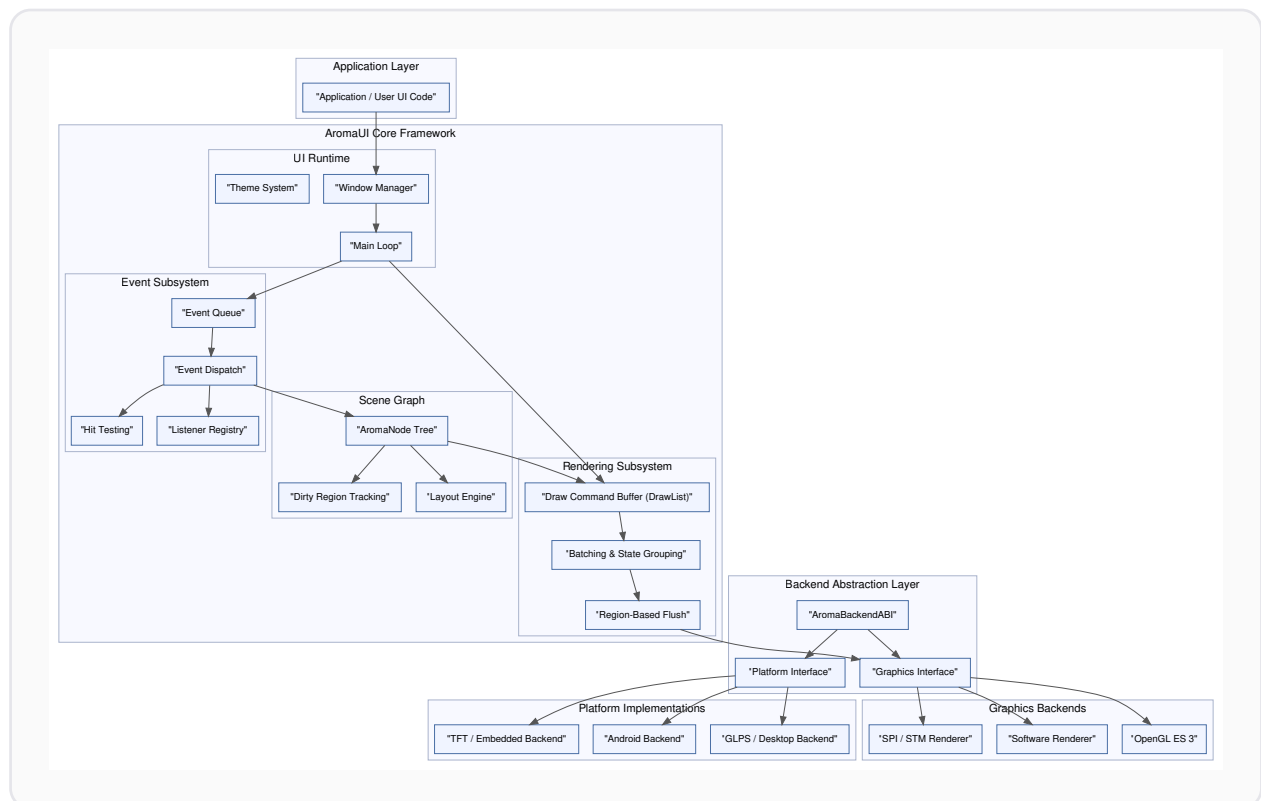
Overview

AromaUI is a modular, cross-platform retained-mode UI framework designed for embedded systems, Android environments, desktop platforms, and software-rendered targets.

The architecture follows a strict layered design to ensure:

- Clear separation of concerns
- Backend independence
- Deterministic memory behavior (embedded-friendly)
- Scalable rendering performance
- Maintainable subsystem boundaries

High-Level Architecture Diagram



Layer Breakdown

1. Application Layer

The application layer contains user-defined UI components and business logic. Applications interact with AromaUI through the Window Manager and Scene Graph APIs.

Responsibilities:

- Create windows and UI trees
- Register event listeners
- Trigger state updates
- Drive application-specific logic

2. AromaUI Core Framework

The core framework is divided into four primary subsystems.

UI Runtime

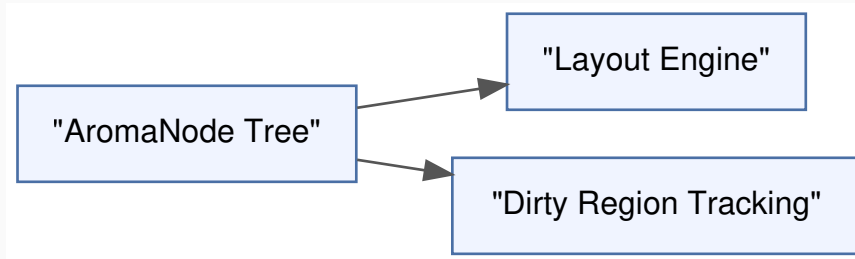
Manages lifecycle and execution.

- Window Manager - controls window instances and root nodes
- Main Loop - drives update and render cycles
- Theme System - centralized styling and visual configuration

Scene Graph

A retained tree of UI nodes.

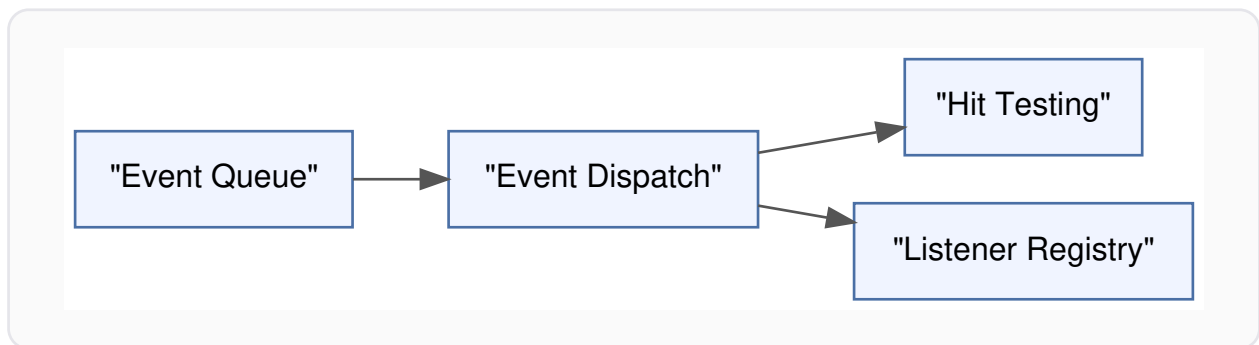
- AromaNode Tree - hierarchical UI structure
- Layout Engine - computes size and position
- Dirty Region Tracking - tracks areas requiring redraw



Event Subsystem

Handles input and dispatch.

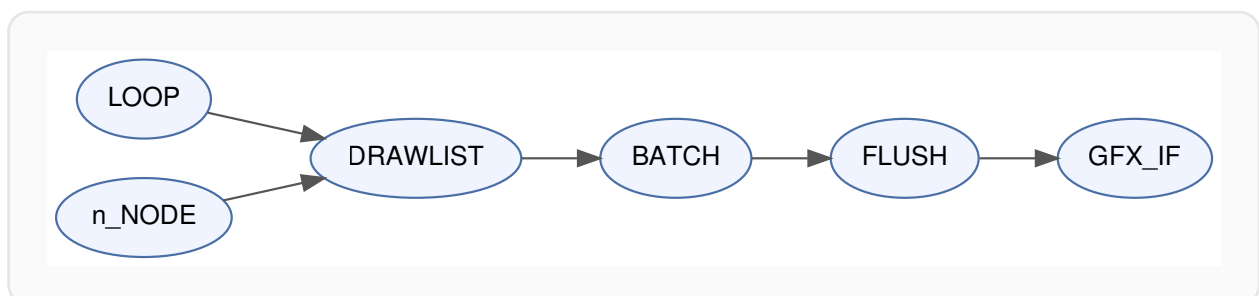
- Event Queue - buffered input events
- Event Dispatch - hierarchical propagation
- Hit Testing - resolves target node
- Listener Registry - efficient listener lookup



Rendering Subsystem

Command-buffer-based rendering pipeline.

- DrawList - records draw commands
- Batching & State Grouping - minimizes backend calls
- Region-Based Flush - partial screen updates



3. Backend Abstraction Layer

Want to learn more about the backend abstraction layer? Check out the [Backend Abstraction Layer](#) documentation for details on how to use it in your code.

Provides strict decoupling between the core engine and platform-specific implementations.

- AromaBackendABI - entry point binding runtime to platform
- Platform Interface - OS/window/input abstraction
- Graphics Interface - rendering abstraction

This design allows the core to remain platform-agnostic.

4. Platform Implementations

Platform-specific bindings provide:

- Window creation
- Input integration
- System lifecycle hooks

Examples:

- Desktop (GLPS)
- Android
- Embedded (TFT/SPI devices)

5. Graphics Backends

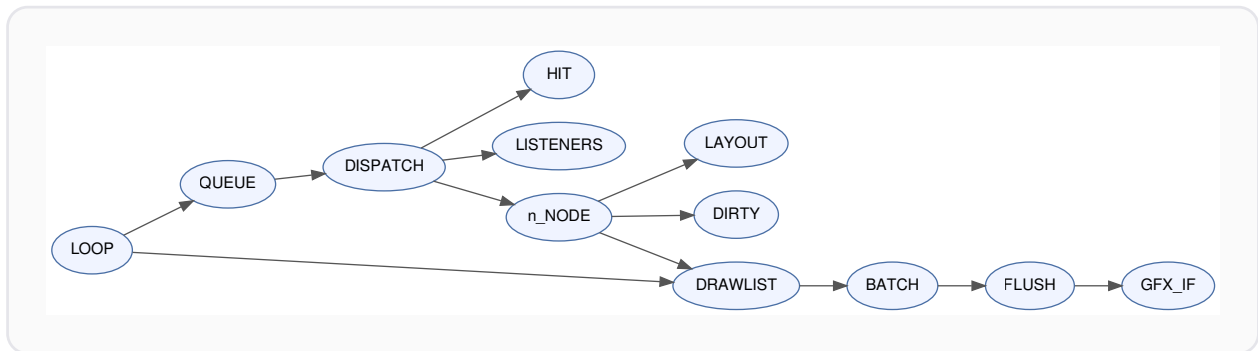
Rendering backends implement the Graphics Interface.

Supported targets:

- OpenGL ES 3
- Vulkan
- TFT/SPI renderer for embedded systems

Each backend receives pre-batched draw commands from the DrawList.

Data Flow Summary



Architectural Principles

- Layered separation of concerns
- Retained-mode scene graph
- Command-buffer rendering
- Backend abstraction
- Embedded-first determinism
- Region-based redraw optimization

Intended Evolution

The architecture supports future expansion including:

- Multi-window compositing
- Threaded rendering pipelines
- GPU resource management layers
- Advanced animation systems
- Custom layout strategies

Core JNI API

Author: AHMED ALI Mohamed Yassine

This section exposes low-level access to the Android runtime. These APIs are intended for advanced use cases such as custom JNI bridging, third-party SDK integration, or manual thread attachment.

All functions are available only when compiling with `__ANDROID__` defined.

```
JNIEnv* aroma_android_get_env();
```

Returns the `JNIEnv` for the current thread.

Returns

- Pointer to `JNIEnv`
- `NULL` if the thread is not attached to the JVM

Example

```
JNIEnv* env = aroma_android_get_env();
if (!env) {
    // Thread not attached
    return;
}
```

Notes

- Each native thread must be attached to the JVM before using JNI.
- Do not store `JNIEnv*` globally across threads.

```
jobject aroma_android_get_activity();
```

Returns a global reference to the active Android `Activity`.

Example

```
jobject activity = aroma_android_get_activity();  
if (activity) {  
    // Use activity in JNI calls  
}
```

Notes

- Managed internally by the platform layer.
- Do not delete this reference.

```
JavaVM* aroma_android_get_jvm();
```

Returns the Java VM instance.

Example

```
JavaVM* jvm = aroma_android_get_jvm();  
if (jvm) {  
    // Attach worker threads if needed  
}
```

Notes

- Useful for background threads requiring JNI access.
- Not required for standard AromaUI usage.

AromaABI Interface

*It is **highly discouraged** to directly interact with the AromaABI in most cases, as it is primarily intended for internal use within the Aroma UI framework. However, understanding the ABI can be beneficial for advanced users who wish to customize or extend the functionality of the framework. The ABI provides a standardized interface for interacting with the graphics and platform backends, allowing for greater flexibility and compatibility across different platforms and graphics APIs.*

Overview

The AromaABI is an interface that defines the functions and types necessary for interacting with the graphics and platform backends of the Aroma UI framework. It allows developers to set and get the types of graphics and platform backends being used, as well as retrieve the corresponding interfaces for each backend. This ABI is essential for ensuring compatibility and proper functioning of the Aroma UI across different platforms and graphics APIs.

Functions

- `void (*set_graphics_backend_type)(AromaGraphicsBackendType type)` : Sets the type of graphics backend to be used by the Aroma UI framework.
- `void (*set_platform_backend_type)(AromaPlatformBackendType type)` : Sets the type of platform backend to be used by the Aroma UI framework.
- `AromaGraphicsBackendType (*get_graphics_backend_type)(void)` : Retrieves the current type of graphics backend being used by the Aroma UI framework.
- `AromaPlatformBackendType (*get_platform_backend_type)(void)` : Retrieves the current type of platform backend being used by the Aroma UI framework.
- `AromaGraphicsInterface* (*get_graphics_interface)(void)` : Retrieves a pointer to the graphics interface corresponding to the current graphics backend.
- `AromaPlatformInterface* (*get_platform_interface)(void)` : Retrieves a pointer to the platform interface corresponding to the current platform backend.

Available Backends

- Graphics Backends:
 - `GRAPHICS_BACKEND_GLES3` : OpenGL ES 3.0 graphics backend.
 - `GRAPHICS_BACKEND_VULKAN` : Vulkan graphics backend.
 - `GRAPHICS_BACKEND_TFT_ESPI` : TFT_eSPI graphics backend for embedded systems.
- Platform Backends:
 - `PLATFORM_BACKEND_GLPS` : GLPS platform backend for desktop applications.
 - `PLATFORM_BACKEND_ANDROID` : Android platform backend for mobile applications.
 - `PLATFORM_BACKEND_TFT_ESPI` : TFT_eSPI platform backend for embedded systems.

Usage

To use the AromaABI, developers can include the `aroma_abi.h` header file in their application code. They can then set the desired graphics and platform backends using the provided functions, and retrieve the corresponding interfaces to interact with the backends. For example:

```
#include "aroma_abi.h"

int main() {
    // Set the graphics and platform backends
    aroma_abi.set_graphics_backend_type(GRAPHICS_BACKEND_VULKAN);
    aroma_abi.set_platform_backend_type(PLATFORM_BACKEND_GLPS);

    // Retrieve the graphics and platform interfaces
    AromaGraphicsInterface* graphics_interface = aroma_abi.get_graphics_interface();
    AromaPlatformInterface* platform_interface = aroma_abi.get_platform_interface();

    // Use the interfaces to perform graphics and platform operations
    // ...

    return 0;
}
```


Backend Compatibility Table

⚠ *Note: The actual implementation of the functions and the behavior of the backends may vary based on the specific platform and graphics API being used. Developers should refer to the documentation for each backend for more details on how to properly utilize them in their applications. It is also discouraged to mix and match graphics and platform backends that are not designed to work together, as this may lead to compatibility issues or unexpected behavior. Always ensure that the chosen graphics and platform backends are compatible with each other and with the target platform for optimal performance and functionality.*

Graphics Backend	Platform Backend	Compatibility
Vulkan	GLPS	Yes
Vulkan	Android	Yes
Vulkan	TFT_eSPI	No
OpenGL ES 3.0	GLPS	Yes
OpenGL ES 3.0	Android	Yes
OpenGL ES 3.0	TFT_eSPI	No
TFT_eSPI	GLPS	No
TFT_eSPI	Android	No
TFT_eSPI	TFT_eSPI	Yes

System Services

Author: AHMED ALI Mohamed Yassine

Provides access to Android system services and application storage paths.

```
jobject aroma_android_get_system_service(const char* service_name);
```

Returns a system service object.

Parameters

- `service_name` – e.g. "vibrator", "wifi", "bluetooth"

Example

```
jobject service = aroma_android_get_system_service("vibrator");
```

Notes

- Returned object is a JNI reference.
- Service availability depends on device and Android version.

```
const char* aroma_android_get_internal_path();
```

Returns the app's internal storage directory.

Example

```
const char* path = aroma_android_get_internal_path();  
printf("Internal: %s\n", path);
```

Notes

- Private to the application.
- No special permissions required.

```
const char* aroma_android_get_external_path();
```

Returns the app's external storage directory.

Example

```
const char* path = aroma_android_get_external_path();
```

Notes

- May require storage permission depending on API level.
- Availability depends on device storage state.

Device Information

Author: AHMED ALI Mohamed Yassine

Provides runtime access to device-level system information.

```
int aroma_android_get_battery_level();
```

Returns the current battery percentage.

Returns

- 0–100
- -1 if unavailable

Example

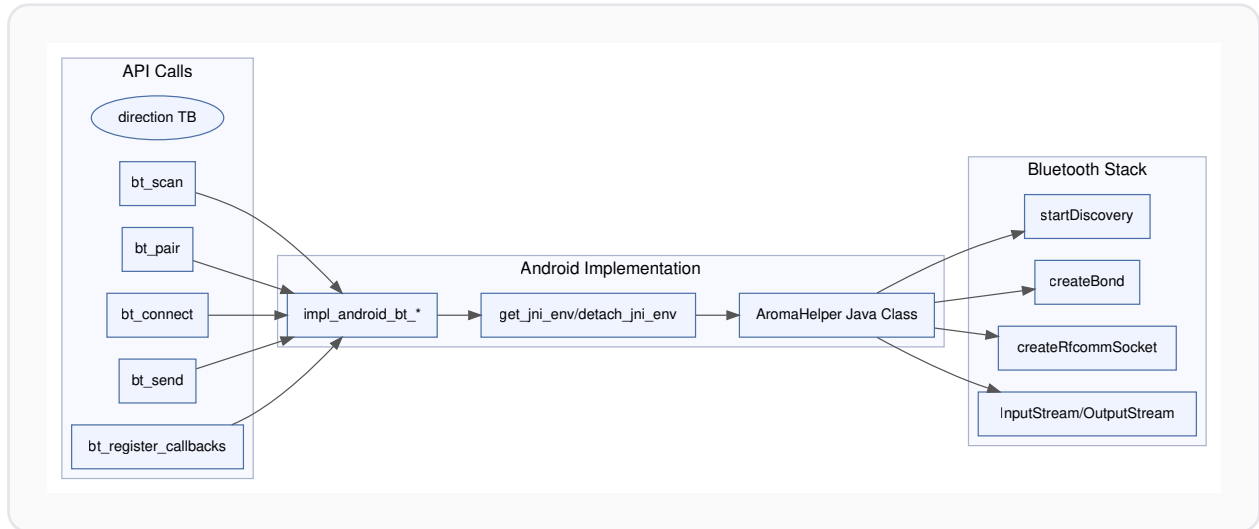
```
int level = aroma_android_get_battery_level();
if (level >= 0) {
    printf("Battery: %d%%\n", level);
}
```

Notes

- May not update in real-time unless refreshed.
- Suitable for dashboards and monitoring apps.

Bluetooth Classic API

Author: AHMED ALI Mohamed Yassine



This guide provides a structured, comprehensive explanation of the AromaUI Bluetooth Classic API for Android.

The API supports:

Feature	Description
Device scanning	Discover nearby Bluetooth Classic devices
Pairing management	Create and remove device bonds
RFCOMM data communication	Send and receive raw data
Connection state management	Monitor and control active connections
Device type detection	Identify connected device category

Limitations:

Limitation	Status
Bluetooth Classic	Supported
Bluetooth Low Energy	Not supported
GATT	Not supported

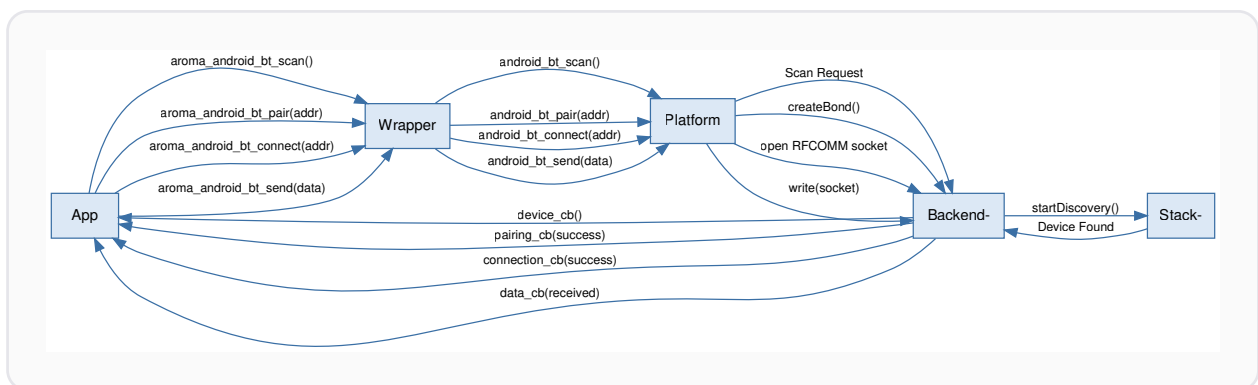
All APIs are valid only when compiling with **ANDROID** defined.

1. Architecture Overview

The Bluetooth layer is exposed through inline wrappers that delegate to AromaPlatformInterface.

The following sequence diagram illustrates a typical Bluetooth workflow:

- Scan for devices
- Pair with device
- Connect using RFCOMM
- Send and receive data



Many operations are asynchronous internally. Design your application around callbacks rather than blocking logic.

2. Android Permissions

Modern Android versions require runtime permissions.

For Android 12 and above:

- android.permission.BLUETOOTH_SCAN
- android.permission.BLUETOOTH_CONNECT
- android.permission.ACCESS_FINE_LOCATION

Check permission:

```
bool granted = aroma_android_check_permission(  
    "android.permission.BLUETOOTH_SCAN"  
);
```

Request permissions:

```
const char* perms[] = {  
    "android.permission.BLUETOOTH_SCAN",  
    "android.permission.BLUETOOTH_CONNECT"  
};  
  
aroma_android_request_permission(perms, 2);
```

Always verify Bluetooth state before scanning:

```
if (!aroma_android_is_bluetooth_enabled()) {  
    // Prompt user to enable Bluetooth  
}
```

3. Registering Callbacks

Register all callbacks before scanning or connecting.

```
aroma_android_bt_register_callbacks(  
    device_cb,  
    scan_finished_cb,  
    pairing_cb,  
    connection_cb,  
    data_cb  
);
```

Callback Signatures

Device discovered:

```
void device_cb(const char* addr,  
               const char* name,  
               int type,  
               int rssi);
```

Scan finished:

```
void scan_finished_cb(void);
```

Pairing result:

```
void pairing_cb(bool success,  
                const char* addr,  
                const char* name);
```

Connection result:

```
void connection_cb(bool success,  
                   const char* addr,  
                   int mode,  
                   int device_type);
```

Data received:

```
void data_cb(const char* data,  
             int len);
```


4. Device Scanning

Scan Modes

Constant	Description
AROMA_BT_SCAN_MODE_PAIRIED	Scan paired devices only
AROMA_BT_SCAN_MODE_NEW	Scan new unpaired devices only
AROMA_BT_SCAN_MODE_ALL	Scan all devices

Start scanning:

```
aroma_android_bt_scan(  
    AROMA_BT_SCAN_MODE_ALL,  
    device_cb  
);
```

Stop scanning:

```
aroma_android_bt_stop_scan();
```

Device callback example:

```
void device_cb(const char* addr,  
               const char* name,  
               int type,  
               int rssi) {  
  
    printf("Device: %s [%s] RSSI: %d Type: %d\n",  
           name, addr, rssi, type);  
}
```

5. Getting Paired Devices

```
char addrs[8][18];  
char names[8][248];  
  
int count = aroma_android_bt_get_paired(  
    addrs,  
    names,  
    8  
);
```

Address format:

XX:XX:XX:XX:XX:XX

Buffers must be preallocated.

6. Pairing Management

Start pairing:

```
bool started = aroma_android_bt_pair(addr);
```

Unpair:

```
aroma_android_bt_unpair(addr);
```

Check bond state:

```
int state = aroma_android_bt_get_pair_state(addr);
```

Bond states:

Constant	Description
AROMA_BT_BOND_NONE	Not bonded
AROMA_BT_BOND_BONDING	Bonding in progress
AROMA_BT_BOND_BONDED	Bonded

7. Connecting to a Device

Default connection:

```
aroma_android_bt_connect(addr);
```

Connection with mode:

```
aroma_android_bt_connect_with_mode(
    addr,
    AROMA_BT_MODE_DATA
);
```

Connection modes:

Constant	Description
AROMA_BT_MODE_DATA	RFCOMM data mode
AROMA_BT_MODE_AUDIO	Audio profile mode
AROMA_BT_MODE_HID	Human Interface Device mode
AROMA_BT_MODE_AUTO	Automatically selected mode

Disconnect:

```
aroma_android_bt_disconnect();
```

Check connection:

```
bool connected = aroma_android_bt_is_connected();
```

8. Sending Data

```
const char* msg = "HELLO";  
  
int sent = aroma_android_bt_send(msg, 5);
```

Returns:

- Number of bytes sent
- Returns negative one on error

Data reception happens through data_cb.

9. Connected Device Information

Device type:

```
int type = aroma_android_bt_get_device_type();
```

Device name:

```
const char* name = aroma_android_bt_get_device_name();
```

Current mode:

```
int mode = aroma_android_bt_get_current_mode();
```

Mode name:

```
const char* modeName = aroma_android_bt_get_mode_name();
```

Wi-Fi API

Author: AHMED ALI Mohamed Yassine

Provides programmatic access to Wi-Fi state and control.

```
bool aroma_android_is_wifi_enabled();
```

Returns whether Wi-Fi is enabled.

Example

```
if (!aroma_android_is_wifi_enabled()) {  
    aroma_android_set_wifi_enabled(true);  
}
```

Notes

- Some Android versions restrict direct Wi-Fi toggling.

```
void aroma_android_set_wifi_enabled(bool enabled);
```

Enables or disables Wi-Fi.

Example

```
aroma_android_set_wifi_enabled(false);
```

Notes

- Behavior varies across Android API levels.
- May require elevated permissions.

UI Utilities

Author: AHMED ALI Mohamed Yassine

Provides lightweight access to common Android UI feedback mechanisms.

```
void aroma_android_toast(const char* msg, bool long_duration);
```

Displays a Toast message.

Example

```
aroma_android_toast("Operation successful", false);
```

Notes

- Non-blocking UI feedback.
- Avoid excessive usage.

```
void aroma_android_open_settings();
```

Opens Android system settings.

Example

```
aroma_android_open_settings();
```

Notes

- Useful when users must manually enable permissions.

```
void aroma_android_vibrate(int ms);
```

Triggers device vibration.

Example

```
aroma_android_vibrate(150);
```

Notes

- Keep vibration short for good UX.

Widgets Library

Author: AHMED ALI Mohamed Yassine

All widget helper functions return:

```
AromaNode*
```

This represents the widget node inserted into the UI tree.

1. Button

Creates a standard clickable button.

```
AromaNode* aroma_ui_button(  
    AromaNode* parent,  
    const char* text,  
    int x, int y, int width, int height,  
    bool (*on_click)(AromaNode*, void*),  
    void* user_data,  
    AromaFont* font  
);
```

Parameters: - parent: Parent node - text: Button label - x, y: Position relative to parent - width, height: Dimensions - on_click: Click callback - user_data: Passed to callback - font: Optional font

2. Button with Icon

```
AromaNode* aroma_ui_button_with_icon(  
    AromaNode* parent,  
    const char* text,  
    int x, int y, int width, int height,  
    bool (*on_click)(AromaNode*, void*),  
    void* user_data,  
    AromaFont* font,  
    const char* icon_code,  
    AromaFont* icon_font  
);
```

Adds an icon to a standard button.

3. Label

```
AromaNode* aroma_ui_label(  
    AromaNode* parent,  
    const char* text,  
    int x, int y,  
    AromaLabelStyle style,  
    AromaFont* font  
);
```

Displays styled text.

4. Container

```
AromaNode* aroma_ui_container(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    AromaLayoutMode layout_mode,  
    AromaFlexDirection flex_dir,  
    AromaJustifyContent justify,  
    AromaAlignItems align  
);
```

Generic layout container supporting flex configuration.

5. Image

From file:

```
AromaNode* aroma_ui_image(  
    AromaNode* parent,  
    const char* path,  
    int x, int y, int width, int height  
);
```

From memory:

```
AromaNode* aroma_ui_image_mem(  
    AromaNode* parent,  
    unsigned char* data,  
    size_t len,  
    int x, int y, int width, int height  
);
```

6. Icon

```
AromaNode* aroma_ui_icon(  
    AromaNode* parent,  
    const char* icon_code,  
    int x, int y, int size,  
    uint32_t color,  
    AromaFont* font  
);
```

Renders an icon using an icon font.

7. Checkbox

```
AromaNode* aroma_ui_checkbox(  
    AromaNode* parent,  
    const char* label,  
    int x, int y, int width, int height,  
    void (*callback)(bool, void*),  
    void* user_data,  
    AromaFont* font  
);
```

8. Radio Button

```
AromaNode* aroma_ui_radiobutton(  
    AromaNode* parent,  
    const char* label,  
    int x, int y, int width, int height,  
    int group_id,  
    void (*callback)(void*),  
    void* user_data,  
    AromaFont* font  
);
```

Grouped selection control.

9. Switch

```
AromaNode* aroma_ui_switch(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    bool initial_state,  
    bool (*on_change)(AromaNode*, void*),  
    void* user_data  
);
```

Boolean toggle control.

10. Slider

```
AromaNode* aroma_ui_slider(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    int min, int max, int value,  
    bool (*on_change)(AromaNode*, void*),  
    void* user_data  
);
```

Numeric range selector.

11. Textbox

```
AromaNode* aroma_ui_textbox(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    const char* placeholder,  
    bool (*on_text_changed)(AromaNode*, const char*, void*),  
    void* user_data,  
    AromaFont* font  
);
```

Text input field.

12. Floating Action Button

```
AromaNode* aroma_ui_fab(  
    AromaNode* parent,  
    int x, int y,  
    AromaFABSize size,  
    const char* icon_text,  
    void (*callback)(void*),  
    void* user_data,  
    AromaFont* font  
);
```

Circular primary action button.

13. Icon Button

```
AromaNode* aroma_ui_iconbutton(  
    AromaNode* parent,  
    const char* icon,  
    int x, int y,  
    int size,  
    AromaIconButtonVariant variant,  
    void (*callback)(void*),  
    void* user_data,  
    AromaFont* font  
);
```

Button that displays only an icon.

14. Chip

```
AromaNode* aroma_ui_chip(  
    AromaNode* parent,  
    const char* label,  
    int x, int y,  
    AromaChipType type,  
    void (*callback)(void*),  
    void* user_data,  
    AromaFont* font  
);
```

With icon:

```
AromaNode* aroma_ui_chip_with_icon(  
    AromaNode* parent,  
    const char* label,  
    int x, int y,  
    AromaChipType type,  
    void (*callback)(void*),  
    void* user_data,  
    AromaFont* font,  
    const char* icon_code,  
    AromaFont* icon_font  
);
```

Compact selectable element.

15. Card

```
AromaNode* aroma_ui_card(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    AromaCardType type  
);
```

Surface container with elevation or outline.

16. Progress Bar

```
AromaNode* aroma_ui_progressbar(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    AromaProgressType type,  
    float progress  
);
```

Determinate or indeterminate progress indicator.

17. Divider

```
AromaNode* aroma_ui_divider(  
    AromaNode* parent,  
    int x, int y, int length,  
    AromaDividerOrientation orientation  
);
```

Horizontal or vertical separator.

18. List View

```
AromaNode* aroma_ui_listview(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    void (*callback)(int, void*),  
    void* user_data,  
    AromaFont* font  
);
```

Scrollable list container.

19. Menu

```
AromaNode* aroma_ui_menu(  
    AromaNode* parent,  
    int x, int y,  
    AromaFont* font  
);
```

Popup or contextual menu.

20. Dialog

```
AromaNode* aroma_ui_dialog(  
    AromaNode* parent,  
    const char* title,  
    const char* message,  
    int width, int height,  
    AromaDialogType type,  
    AromaFont* font  
);
```

Modal dialog component.

21. Tabs

```
AromaNode* aroma_ui_tabs(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    const char** labels, int count,  
    void (*on_change)(AromaNode*, int, void*),  
    void* user_data,  
    AromaFont* font  
);
```

With icons:

```
AromaNode* aroma_ui_tabs_with_icons(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    const char** labels,  
    const char** icons,  
    int count,  
    void (*on_change)(AromaNode*, int, void*),  
    void* user_data,  
    AromaFont* font,  
    AromaFont* icon_font  
);
```

22. Snackbar

```
AromaNode* aroma_ui_snackbar(  
    AromaNode* parent,  
    const char* message,  
    int duration_ms,  
    AromaFont* font  
);
```

Temporary bottom notification.

23. Tooltip

```
AromaNode* aroma_ui_tooltip(  
    AromaNode* parent,  
    const char* text,  
    int x, int y,  
    AromaTooltipPosition pos,  
    AromaFont* font  
);
```

Hover hint element.

24. Sidebar

```
AromaNode* aroma_ui_sidebar(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    const char** labels, int count,  
    void (*on_select)(AromaNode*, int, void*),  
    void* user_data,  
    AromaFont* font  
);
```

With icons:

```
AromaNode* aroma_ui_sidebar_with_icons(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    const char** labels,  
    const char** icons,  
    int count,  
    void (*on_select)(AromaNode*, int, void*),  
    void* user_data,  
    AromaFont* font,  
    AromaFont* icon_font  
);
```

25. Dropdown

```
AromaNode* aroma_ui_dropdown(  
    AromaNode* parent,  
    int x, int y, int width, int height,  
    char** options,  
    int option_count,  
    void (*on_selection_changed)(int, const char*, void*),  
    void* user_data,  
    AromaFont* font  
);
```

Selectable list popup.

26. Debug Overlay

```
AromaNode* aroma_ui_debug_overlay(  
    AromaNode* parent,  
    int x, int y, int width,  
    AromaFont* font  
);
```

Displays runtime diagnostics.

27. Window Node

```
AromaNode* aroma_ui_window(  
    const char* title,  
    int width, int height,  
    bool fullscreen  
);
```

Creates a top level window node.

Layout Management

Layout Management

AromaUI provides flexible layout management for responsive UIs. Layout containers support flexbox-like configuration and adaptive sizing.

Features

- Flex and grid layouts
- Responsive sizing
- Alignment and justification
- Nested containers

Architecture

```
flowchart TD
    Window[Window] --> Container[Container]
    Container --> Widget[Widget]
    Container --> Container2[Container2[Nested Container]]
```

Example Usage

```
AromaNode* root = aroma_ui_window("App", 800, 600, false);
AromaNode* container = aroma_ui_container(root, 0, 0, 800, 600, FLEX, ROW, CENTER, START);
AromaNode* button = aroma_ui_button(container, "Click", 10, 10, 100, 40, on_click, NULL, font);
```

API Reference

- `aroma_ui_container(...)`
- `aroma_ui_window(...)`

DPI Conversion

Author: AHMED ALI Mohamed Yassine

1. Overview

The AromaUI DPI System provides density-aware utilities for Android builds.

It ensures consistent layout scaling across different screen densities, physical sizes, and user font preferences.

This subsystem allows you to:

- Query density metrics
- Convert between DP, PX, and SP units
- Detect physical screen size
- Retrieve available window dimensions
- Respect user font scaling
- Access true hardware DPI values

All APIs are valid **only when compiled with** `__ANDROID__` **defined.**

2. Core Density Concepts

Density (Scale Factor)

Density represents logical scaling relative to the baseline **160 DPI (mdpi)**.

DPI Category	Density	Example DPI
mdpi	1.0	160
hdpi	1.5	240
xhdpi	2.0	320
xxhdpi	3.0	480
xxxhdpi	4.0	640

Retrieve density scale factor:

```
float density = aroma_android_get_density();
```

Retrieve raw density DPI:

```
int dpi = aroma_android_get_density_dpi();
```

3. Text Scaling (Scaled Density)

Scaled density accounts for the user's font size setting.

Use for text scaling (SP units).

```
float scaledDensity = aroma_android_get_scaled_density();
```

Never use plain density for text scaling.

4. Unit Conversions

DP -> PX

Density-independent pixels to physical pixels.

```
int px = aroma_android_dp_to_px(16);
```

Conceptual formula:

$$px = dp \times density$$

PX -> DP

```
int dp = aroma_android_px_to_dp(48);
```

SP -> PX (Text)

```
int px = aroma_android_sp_to_px(14);
```

Conceptual formula:

$$px = sp \times scaledDensity$$

PX -> SP

```
int sp = aroma_android_px_to_sp(28);
```

5. Window Size (DP Units)

Retrieve available layout space excluding system UI (status bar, navigation bar).

```
int width_dp;
int height_dp;

aroma_android_get_available_size_dp(&width_dp, &height_dp);
```

This should be used for responsive layout logic instead of raw pixel values.

6. Physical Screen Size

Width and Height in Inches

```
float width_in;  
float height_in;  
  
aroma_android_get_screen_size_inches(&width_in, &height_in);
```

Diagonal Size in Inches

```
float diagonal = aroma_android_get_screen_diagonal_inches();
```

Useful for differentiating:

- Phone
- Tablet
- Large tablet
- Embedded display

7. Screen Size Category

Returns a logical size classification string:

```
const char* category = aroma_android_get_screen_size_category();
```

Possible values:

Value	Meaning
small	Small phone
normal	Standard phone
large	Phablet / small tab
xlarge	Tablet
xxlarge	Large tablet / TV

8. True Physical DPI

Retrieve hardware DPI along each axis:

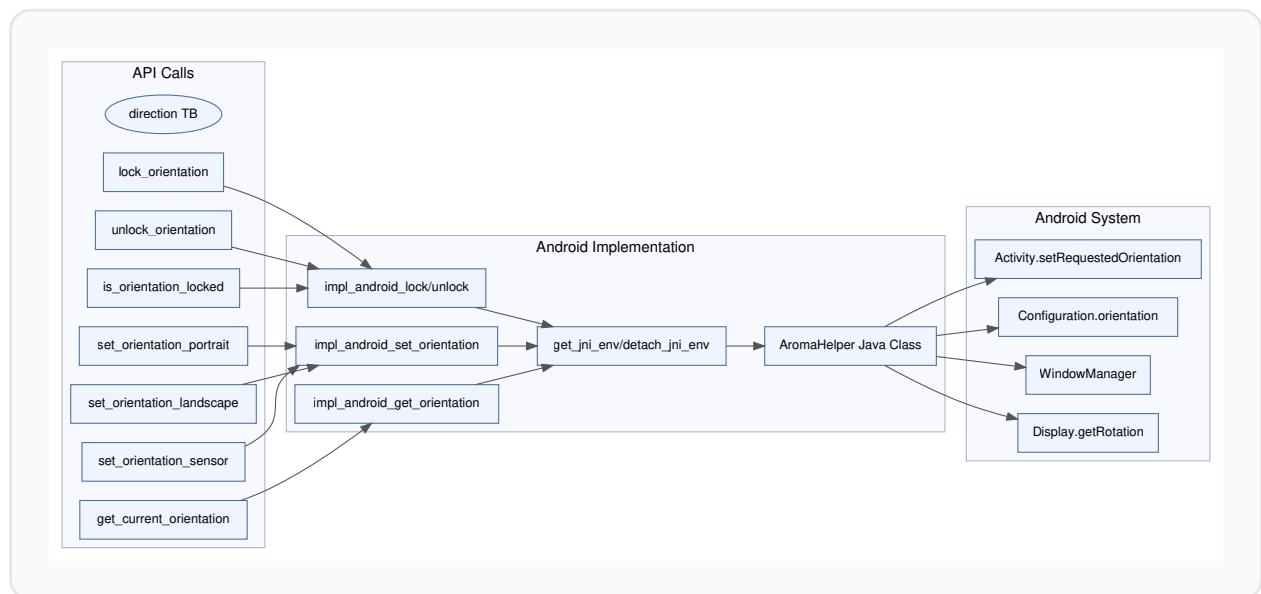
```
float xdpi = aroma_android_get_xdpi();  
float ydpi = aroma_android_get_ydpi();
```

These values represent the actual physical dots-per-inch of the panel and may differ from logical density.

Orientation API

Author: AHMED ALI Mohamed Yassine

This guide provides a structured, comprehensive explanation of the AromaUI Screen Orientation Control API for Android.



API Capabilities

Feature	Description
Orientation locking	Prevent screen rotation
Orientation unlocking	Allow sensor-based rotation
Force portrait	Lock screen to portrait mode
Force landscape	Lock screen to landscape mode
Sensor-based rotation	Enable automatic rotation
Current orientation query	Get device orientation state
Orientation lock status	Check if orientation is locked

Limitations

Limitation	Status
Portrait mode	Supported
Landscape mode	Supported
Reverse portrait	Not directly
Reverse landscape	Not directly
Sensor-based	Supported
User-sensor	Not directly

⚠ All APIs are valid only when compiling with `ANDROID` defined.

1. Android Manifest Configuration

To use orientation control, ensure your Android manifest includes appropriate configuration:

```
<activity
    android:name=".MainActivity"
    android:configChanges="orientation|screenSize"
    android:screenOrientation="unspecified">
</activity>
```

The `configChanges` attribute allows your app to handle orientation changes manually rather than letting Android recreate the activity.

2. Basic Orientation Control

Lock Current Orientation

Prevent the screen from rotating based on sensor input:

```
aroma_android_lock_orientation();
```

This locks the screen to whatever orientation the device is currently in (portrait or landscape).

Unlock Orientation

Allow the screen to auto-rotate based on device sensors:

```
aroma_android_unlock_orientation();
```

This returns the device to sensor-based rotation behavior.

3. Forcing Specific Orientations

Force Portrait Mode

Lock the screen to portrait orientation regardless of device position:

```
aroma_android_set_orientation_portrait();
```

This forces the display to portrait mode (taller than wide).

Force Landscape Mode

Lock the screen to landscape orientation regardless of device position:

```
aroma_android_set_orientation_landscape();
```

This forces the display to landscape mode (wider than tall).

Enable Sensor-Based Rotation

Set the orientation to follow device sensors:

```
aroma_android_set_orientation_sensor();
```

This is equivalent to unlocking orientation but explicitly sets sensor mode.

4. Querying Orientation State

Get Current Orientation

Retrieve the current screen orientation:

```
int orientation = aroma_android_get_current_orientation();

if (orientation == 1) {
    // Portrait mode
} else if (orientation == 2) {
    // Landscape mode
}
```

Return Values

Value	Description
1	Portrait orientation
2	Landscape orientation
-1	Unknown/unavailable

Check if Orientation is Locked

Determine if orientation locking is currently active:

```
bool is_locked = aroma_android_is_orientation_locked();

if (is_locked) {
    // Orientation is locked to a specific mode
} else {
    // Orientation can auto-rotate
}
```

5. Complete Usage Example


```

#include "aroma_android.h"

// Example: Toggle between portrait and landscape
void toggle_orientation(void) {
    int current = aroma_android_get_current_orientation();

    if (current == 1) {
        // Currently in portrait, switch to landscape
        aroma_android_set_orientation_landscape();
        printf("Switched to landscape mode\n");
    } else if (current == 2) {
        // Currently in landscape, switch to portrait
        aroma_android_set_orientation_portrait();
        printf("Switched to portrait mode\n");
    }
}

// Example: Lock/unlock orientation
void handle_orientation_lock(bool should_lock) {
    if (should_lock) {
        aroma_android_lock_orientation();
        printf("Orientation locked\n");
    } else {
        aroma_android_unlock_orientation();
        printf("Orientation unlocked\n");
    }
}

// Example: Initialize with desired orientation
void initialize_with_orientation(void) {
    // Force portrait mode at startup
    aroma_android_set_orientation_portrait();

    // Later, allow rotation
    // aroma_android_set_orientation_sensor();
}

// Example: Respond to device state
void on_device_rotated(void) {
    if (!aroma_android_is_orientation_locked()) {
        int orientation = aroma_android_get_current_orientation();

        switch(orientation) {
            case 1:
                // Adapt UI for portrait
                break;

```

```
        case 2:
            // Adapt UI for landscape
            break;
    }
}
```

6. Best Practices

When to Lock Orientation

Use Case	Recommendation
Video playback	Lock to landscape
Document reading	Allow auto-rotation
Games	Lock to preferred orientation
Camera apps	Lock to landscape
E-books	Allow portrait/landscape
Media viewers	Follow sensor

Handling Configuration Changes

When your app handles orientation changes manually:

```
// In your Android activity's onCreate or native code
aroma_android_set_orientation_portrait();

// The layout will update through aroma_refresh_layout()
```

Performance Considerations

- Orientation changes trigger surface recreation
- Layout recalculation occurs automatically
- Consider debouncing rapid orientation changes
- Cache orientation-dependent resources

7. Error Handling

```
// Always check if orientation functions are available
if (!aroma_android_get_current_orientation) {
    printf("Orientation API not available\n");
    return;
}

// Check return values where applicable
int orientation = aroma_android_get_current_orientation();
if (orientation < 0) {
    printf("Failed to get orientation\n");
}

// Verify lock state before assuming
if (aroma_android_is_orientation_locked()) {
    // Safe to assume fixed orientation
}
```

Theming System

A Baked-In set of themes is included in AromaUI, which can be used as-is or customized to fit the needs of your application. The available themes include:

Theme Name	Description
Default Theme	A balanced default theme suitable for most applications.
High Contrast Theme	A theme with high contrast colors for improved readability and accessibility.
Material Theme Presets	A set of themes based on the Material Design color palette, including presets for purple, blue, teal, green, orange, and pink.
Black OLED Theme	A theme with darker colors for a more subdued and modern look, suitable for low-light environments.
Material Dark Theme Presets	A set of dark themes based on the Material Design color palette, including presets for purple, blue, teal, green, orange, and pink.

Customization

In addition to the built-in themes, developers can create their own custom themes by defining their own color palettes, spacing, typography, and other style properties.

- The `aroma_theme_create_custom` function allows developers to create a custom theme by providing their own values for the various theme properties.
- Developers can also modify the global theme using the `aroma_theme_set_global` function, which will affect all components that use the global theme for styling.
- The `aroma_style_apply_theme_colors` function can be used to apply the colors from a theme to a specific style, allowing for more granular control over the appearance of individual components.

Color Manipulation

The theming system also includes utility functions for manipulating colors, such as adjusting brightness, blending colors, and converting between different color formats. These functions can be useful for creating dynamic themes or for implementing features such as dark mode or user-customizable themes.

For example, the `aroma_color_adjust` function can be used to adjust the brightness of a color by a specified factor, while the `aroma_color_blend` function can be used to blend two colors together based on a specified blend factor. The `aroma_color_rgb` and `aroma_color_rgba` functions can be used to create colors from RGB or RGBA values, while the `aroma_color_extract_rgb` function can be used to extract the RGB components from a color value.

Shadows

The theming system also includes support for shadows, which can be used to add depth and visual interest to UI components. - The `AromaShadow` struct defines the properties of a shadow, including the blur radius, offset, color, and opacity.

- The `aroma_shadow_create_*` functions provide predefined shadow styles, while the `aroma_shadow_create_custom` function allows developers to create their own custom shadow styles.
- The `aroma_style_apply_shadow` function can be used to apply a shadow to a specific style, allowing for more control over the appearance of individual components. Shadows can be particularly effective when used in combination with themes, as they can help to create a sense of hierarchy and focus within the UI.

For example, a component with a deeper shadow may be perceived as being more important or interactive than a component with a softer shadow. Additionally, shadows can be used to create a sense of depth and layering within the UI, which can enhance the overall visual appeal and user experience of the application.

Gestures & Touch

Gestures & Touch

Handle touch events, gestures, and multi-touch interactions in AromaUI.

Features

- Tap, swipe, pinch, drag
- Multi-touch support
- Gesture callbacks

Architecture

```
flowchart TD
    Touch[Touch Event] --> Recognize[Gesture Recognizer]
    Recognize --> Callback[Callback]
```

Example Usage

```
aroma_ui_set_gesture(widget, GESTURE_SWIPE, on_swipe);
```

API Reference

- `aroma_ui_set_gesture(widget, gesture, callback)`

Camera API

Author: AHMED ALI Mohamed Yassine

Provides access to Android media intents for capturing and selecting images.

```
void aroma_android_launch_camera();
```

Launches the default camera application.

Example

```
aroma_android_launch_camera();
```

Notes

- Requires camera permission.
- Result must be handled on the Java side.

```
void aroma_android_launch_gallery();
```

Launches image picker.

Example

```
aroma_android_launch_gallery();
```

Notes

- Used for selecting existing media.
- Result is delivered asynchronously.

File System API

File System API

AromaUI provides cross-platform file and directory operations for Android, Linux, and Windows.

Features

- File read/write
- Directory management
- File metadata

Architecture

```
flowchart TD
    App[Application] --> FS[File System]
    FS --> Read[Read/Write]
    FS --> Dir[Directory Management]
    FS --> Meta[Metadata]
```

Example Usage

```
aroma_fs_read("file.txt");
aroma_fs_write("file.txt", data);
aroma_fs_list_dir("/home/user");
```

API Reference

- `aroma_fs_read(path)`
- `aroma_fs_write(path, data)`
- `aroma_fs_list_dir(path)`

Preferences API

Preferences API

Simple key-value storage for user preferences and settings.

Features

- Store/retrieve preferences
- Data persistence
- Type support (int, string, bool)

Architecture

```
flowchart TD
    App[Application] --> Prefs[Preferences]
    Prefs --> Store[Store Value]
    Prefs --> Retrieve[Retrieve Value]
```

Example Usage

```
aroma_prefs_set("theme", "dark");
aroma_prefs_get("theme");
```

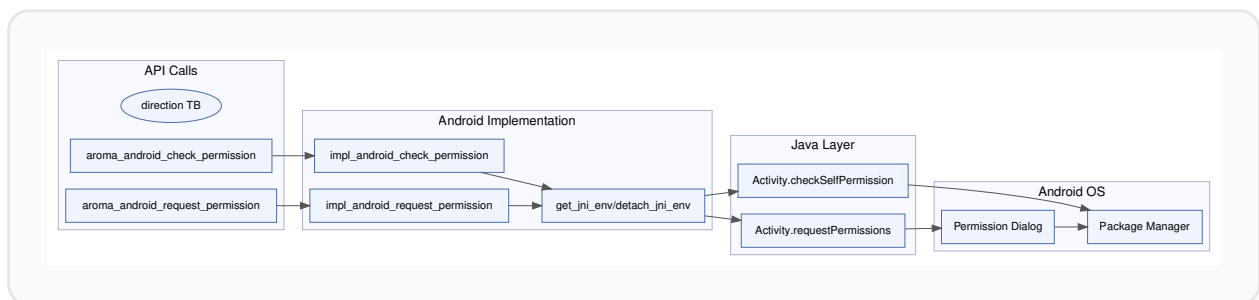
API Reference

- `aroma_prefs_set(key, value)`
- `aroma_prefs_get(key)`

Permissions API

Author: AHMED ALI Mohamed Yassine

The Permissions API provides runtime permission management in compliance with Android's security model. It supports permission grouping, Android version–specific behavior, and asynchronous permission handling.



1. Permission Checking

```
bool aroma_android_check_permission(const char* permission_name);
```

Checks whether a specific permission has been granted to the application.

Parameters

Parameter	Description	Example
<code>permission_name</code>	Fully qualified Android permission string	<code>"android.permission.CAMERA"</code>

Returns

Value	Description
<code>true</code>	Permission is granted
<code>false</code>	Permission is denied or not yet requested

Example

```
if (!aroma_android_check_permission("android.permission.CAMERA")) {  
    const char* permissions[] = {"android.permission.CAMERA"};  
    aroma_android_request_permissions(permissions, 1);  
}
```

2. Permission Requesting

```
void aroma_android_request_permissions(const char** permissions, int permCount);
```

Requests one or more permissions from the user.

Parameters

Parameter	Description	Example
<code>permissions</code>	Array of permission strings	<code>{ "android.permission.CAMERA", "android.permission.RECORD_AUDIO" }</code>
<code>permCount</code>	Number of permissions in the array	<code>2</code>

Example

```
// Request camera and microphone permissions
const char* permissions[] = {
    "android.permission.CAMERA",
    "android.permission.RECORD_AUDIO"
};

aroma_android_request_permissions(permissions, 2);
```

3. Android Version Behavior

Android Version	Behavior
API < 23	Permissions granted at install time. Function returns immediately.
API 23+	Runtime permission dialog(s) shown to user.
API 31+	Automatically handles Bluetooth–Location dependency.

4. Automatic Permission Grouping

Permissions are grouped to comply with Android system requirements.

Group	Permissions
NEARBY_DEVICES	BLUETOOTH_SCAN, BLUETOOTH_ADVERTISE, BLUETOOTH_CONNECT
LOCATION	ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION, ACCESS_BACKGROUND_LOCATION
NOTIFICATIONS	POST_NOTIFICATIONS
STORAGE	READ_EXTERNAL_STORAGE, WRITE_EXTERNAL_STORAGE, READ_MEDIA_IMAGES, READ_MEDIA_VIDEO, READ_MEDIA_AUDIO
CAMERA	CAMERA
MICROPHONE	RECORD_AUDIO

5. Common Permission Strings

Permission	String Constant
Camera	android.permission.CAMERA
Fine Location	android.permission.ACCESS_FINE_LOCATION
Coarse Location	android.permission.ACCESS_COARSE_LOCATION
Bluetooth Scan	android.permission.BLUETOOTH_SCAN
Bluetooth Connect	android.permission.BLUETOOTH_CONNECT
Record Audio	android.permission.RECORD_AUDIO
Read Storage	android.permission.READ_EXTERNAL_STORAGE
Write Storage	android.permission.WRITE_EXTERNAL_STORAGE
Post Notifications	android.permission.POST_NOTIFICATIONS

Building & Deployment

Author: AHMED ALI Mohamed Yassine

This document describes the complete deployment workflow for AromaUI projects using the `aroma` CLI. It focuses strictly on building, signing, packaging, and distributing applications for Linux and Android.

1. Build Targets Overview

AromaUI supports deployment to:

- Linux (native executable)
- Android Debug APK (testing)
- Android Release APK (signed, production)
- Android App Bundle (AAB, Play Store)

All deployment commands must be executed from the root of your Aroma project.

Linux Deployment

Build Linux Binary

```
aroma build linux
```

This command:

- Creates a `build/` directory
- Runs CMake configuration
- Compiles using Make

Output executable:

```
build/<project_name>
```

Run Linux Build

```
aroma run linux
```

This builds (if needed) and launches the executable.

For distribution, package the binary along with any required shared libraries.

Android Deployment

Android deployment supports debug and release builds.

Debug Build (Testing)

Build Debug APK

```
aroma build android
```

Output:

```
android/app/build/outputs/apk/debug/app-debug.apk
```

Install on Connected Device

```
aroma run android
```

Requirements:

- USB debugging enabled
- `adb devices` shows a connected device

Run in Emulator

```
aroma run android --emu
```

This will:

- Create an AVD if needed
- Start the emulator
- Wait for boot completion
- Install debug APK
- Launch the application

Release Deployment (Production)

Release builds require a signing keystore.

Step 1: Configure Signing

Run:

```
aroma sign
```

This process:

- Creates `android/keystore/release.keystore`
- Generates `android/keystore.properties`
- Updates `app/build.gradle` with signing configuration

The signing configuration is automatically connected to the Gradle `release` build type.

Important:

- Keep the keystore file secure
- Do not lose the password
- Do not commit the keystore to public repositories

Step 2: Build Release APK

```
aroma build android --release
```

Signed APK output:

```
android/app/build/outputs/apk/release/app-release.apk
```

If unsigned, the output will show:

```
app-release-unsigned.apk
```

Unsigned APKs cannot be installed or distributed.

Step 3: Build Android App Bundle (AAB)

For Google Play submission:

```
aroma build android --release --aab
```

Output:

```
android/app/build/outputs/bundle/release/app-release.aab
```

The AAB format is required for Play Store uploads.

Manual Installation of Release APK

To install a signed release APK manually:

```
adb install -r android/app/build/outputs/apk/release/app-release.apk
```

If upgrading an existing version, ensure the keystore matches the previously installed version.

Continuous Deployment Recommendations

For production workflows:

- Store keystore securely (offline backup)
- Use CI to run:

```
aroma build android --release --aab
```

- Archive generated AABs per version
- Tag releases in version control

Play Store Deployment Checklist

Before uploading to Google Play:

- Version code incremented
- Version name updated
- Release build signed
- AAB generated
- App tested on physical device
- Debug logs removed
- Min SDK and Target SDK verified

Upload file:

```
app-release.aab
```

Troubleshooting Deployment

Keystore Password Incorrect

Verify with:

```
keytool -list -keystore release.keystore
```

If the password is wrong, you must recreate signing and cannot update existing published apps.

No Device Detected

Check:

```
adb devices
```

Restart adb if needed:

```
adb kill-server  
adb start-server
```

SDK Not Found

Run:

```
aroma doctor
```

Or install manually:

```
aroma install-sdk
```

Deployment Command Summary

Debug APK:

```
aroma build android
```

Release APK:

```
aroma build android --release
```

Release AAB:

```
aroma build android --release --aab
```

Run on device:

```
aroma run android
```

Run in emulator:

```
aroma run android --emu
```

Linux build:

```
aroma build linux
```

Linux run:

```
aroma run linux
```

Logging System

The logging system in AromaUI is designed to provide developers with a flexible and efficient way to log messages, errors, and other information during the development and debugging process. The logging system supports multiple log levels, allowing developers to categorize their log messages based on their importance and severity. The available log levels include:

Log Level	Description
DEBUG_LEVEL_INFO	General information about the application's state.
DEBUG_LEVEL_WARNING	Indications of potential issues or important events.
DEBUG_LEVEL_ERROR	Errors that have occurred in the application.
DEBUG_LEVEL_CRITICAL	Severe errors that may cause the application to crash.

*Note: You can set a minimum log level to filter out less important messages. For example, setting the log level to **WARNING** will only log **WARNING**, **ERROR**, and **CRITICAL** messages, while **INFO** messages will be ignored. `set_minimum_log_level(LogLevel level)` can be used to set the minimum log level for the logging system. You can also enable or disable logging `set_logging_enabled(bool enabled);`.*

Logging Functions

The logging system provides a set of functions that developers can use to log messages at different log levels.

Function Name	Description
<code>LOG_INFO(const char* format, ...)</code>	Logs an informational message.
<code>LOG_WARNING(const char* format, ...)</code>	Logs a warning message.
<code>LOG_ERROR(const char* format, ...)</code>	Logs an error message.
<code>LOG_CRITICAL(const char* format, ...)</code>	Logs a critical error message.

Performance & Metrics

The logging system also includes a timer utility that allows developers to measure the time taken for specific operations or code blocks. This can be useful for performance profiling and optimization. The timer utility provides functions to start, stop, and reset timers, as well as retrieve the elapsed time in milliseconds.

Function Name	Description
<code>LOG_PERFORMANCE(NULL)</code>	Starts a performance timer.
<code>LOG_PERFORMANCE("Timer Name")</code>	Ends the specified performance timer.

Memory Debugging

In addition to logging messages and performance metrics, the logging system also includes a memory dump utility that allows developers to dump the contents of memory buffers for debugging purposes. This can be particularly useful for diagnosing issues related to memory management, such as buffer overflows or memory leaks. The memory dump utility provides a function to dump the contents of a specified memory buffer in a readable format.

Function Name	Description
<code>void dump_memory(const char *label, const void *buffer, size_t size);</code>	Dumps the contents of the specified memory buffer with a label for identification.

Exporting Logs

The logging system also provides functionality to export logs to a file for later analysis. This can be useful for debugging issues that occur in production environments or for keeping a record of application events. The log export function allows developers to specify the file path and name for the exported log file.

Function Name	Description
<code>void save_log_file(const char *path);</code>	Saves the current log messages to a file at the specified path.

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